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Cite this article: Ebrahimi S, Bagchi P. 2022

Application of machine learning in predicting blood flow and red cell distribution in capillary vessel networks. *J. R. Soc. Interface* **19**:

20220306.

<https://doi.org/10.1098/rsif.2022.0306>

Received: 19 April 2022

Accepted: 21 July 2022

Subject Category:

Life Sciences—Engineering interface

Subject Areas:

biomedical engineering, biomechanics, computational biology

Keywords:

microcirculation, computational fluid dynamics, blood cell, haemodynamics, machine learning

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Electronic supplementary material is available online at <https://doi.org/10.6084/m9.figshare.c.6125228>.

Application of machine learning in predicting blood flow and red cell distribution in capillary vessel networks

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Capillary blood vessels in the body partake in the exchange of gas and nutrients with tissues. They are interconnected via multiple vascular junctions forming the microvascular network. Distributions of blood flow and red cells (RBCs) in such networks are spatially uneven and vary in time. Since they dictate the pathophysiology of tissues, their knowledge is important. Theoretical models used to obtain flow and RBC distribution in large networks have limitations as they treat each vessel as a one-dimensional segment and do not explicitly consider cell–cell and cell–vessel interactions. High-fidelity computational models that accurately model each individual RBC are computationally too expensive to predict haemodynamics in large vascular networks and over a long time. Here we investigate the applicability of machine learning (ML) techniques to predict blood flow and RBC distributions in physiologically realistic vascular networks. We acquire data from high-fidelity simulations of deformable RBC suspension flowing in the networks. With the flow and haematocrit specified at an inlet of vasculature, the ML models predict the time-averaged flow rate and RBC distributions in the entire network, time-dependent flow rate and haematocrit in each vessel and vascular bifurcation in isolation over a long time, and finally, simultaneous spatially and temporally evolving quantities through the vessel hierarchy in the networks.

1. Introduction

The primary function of the circulatory system is to supply oxygen and nutrients to tissues and scavenge cellular waste products. Red blood cells (RBCs), which constitute nearly 45% of the blood volume, serve as the oxygen carrier. The exchange of oxygen and other biochemicals between blood and tissues primarily occurs in the smallest blood vessels, known as capillary vessels. Together with their upstream and downstream vessels known as arterioles and venules, and vascular junctions, they form architecturally complex vascular networks, also referred to as microvascular networks [1–5]. These vessels also play a significant role in many diseases, in addition to serving as exchange vessels, e.g. in heart and cerebral disorders, diabetes, tumour growth, sickle cell anaemia and malaria [6–9].

Under healthy conditions, vessels in a microvascular network are arranged to maintain tissue perfusion and, hence, oxygen delivery. Yet, *in vivo* studies have shown that the distribution of blood flow and RBCs in a network is both spatially and temporally heterogeneous [10–14]. The distribution of blood and RBCs in a microvascular network is critical to healthy functioning as well as disease progression [15–21]. For example, capillary vessel regression may occur due to lack of blood flow, while angiogenesis may be triggered by tissue hypoxia. Therefore, the knowledge of how RBCs and blood flow are distributed in a microvascular network is of immense pathophysiological importance.

Although recent advances in imaging techniques have enabled mapping of a large number of capillary vessels *in vivo* [14,22,23], direct measurements of spatially and temporally resolved RBC and flow distribution are possible only over relatively small tissue volumes [24]. To overcome this limitation, theoretical models of network blood flow are often used to obtain flow and RBC distribution in a large microvascular network after image acquisition [23,25,26]. Such theoretical models, although used widely, have some limitations. For example, they treat each blood vessel as a one-dimensional segment thereby neglecting the influence of vessel curvature. Further, they do not consider cell–cell and cell–vessel interactions that can affect cell distribution and cause time-dependent fluctuations in blood flow. Additionally, such models use empirical relations to prescribe blood viscosity as well as the way RBCs distribute at vascular junctions. Recent studies have shown that RBC–vessel and RBC–RBC interaction, and RBC deformability affect RBC partitioning at bifurcations and hence their distribution throughout the vascular network [27–33].

High-fidelity computational models that treat blood as a suspension of individual deformable cells have emerged over the past two decades, e.g. [31,34–41], to name a few. Such models can accurately predict cell–cell and cell–vessel interactions and have been used to elucidate the microhaemodynamics of healthy and diseased blood cells, although in simple geometry such as flow through straight channels and circular tubes. Recently, such models have been extended to predict cell dynamics in networks of multiple, interconnected capillary vessels and bifurcations that are geometrically complex but represent physiologically realistic network architecture [32,33,42–45]. Although these simulations provide a wealth of information in terms of microscale haemodynamics at sub-cellular resolutions, they are, however, time consuming and are limited to a small tissue area and a short physical flow time. Therefore, such cell-resolved models may not be useful when larger tissue areas and several cardiac cycles need to be considered.

As a first step towards overcoming such limitations, here we investigate whether machine learning (ML) techniques can be used to predict flow and RBC distributions in microvascular networks over longer times. ML has been successfully applied in many areas from computer vision to speech recognition, drug discovery etc. [46]. It is being increasingly used in biomedical and fluid mechanical applications [47,48]. Examples include detection of diabetic retinopathy, cancer, and vascular diseases [49–51], prediction of turbulent flow, flow around airfoils and bluff bodies, flow in porous media, multiphase flows, fluid–structure interaction [48] and modelling blood flow in the microcirculation [52,53]. ML techniques are also applied to microscale blood flow. Examples include classification of RBC shapes in capillary blood flow [54], modelling RBC deformation and prediction of RBC trajectory in microfluidic devices [55,56], and estimation of cell deformability [57,58]. ML was also used to integrate images of blood flow to underlying physical laws to infer the flow field in microaneurysm [59]. Additionally, combined ML and high-fidelity simulation was considered to study the flow of vesicle suspension to reduce expensive computation [60].

Here we investigate the applicability of ML techniques to predict blood flow and RBC distribution in physiologically realistic and large microvascular networks over long times. Both time-averaged distribution and time-dependent variations are predicted. The methodology, which includes

specific ML models and generating training data, is described in the next section. Thereafter, ML-predicted haemodynamic results are presented, first for the time-averaged but spatially dependent quantities followed by time series predictions, and finally for simultaneous spatial and temporal variations.

2. Methodology

2.1. Data acquisition

As is the case for any ML application, the first step is to acquire the necessary data for developing and testing the ML models. To generate the data, we use high-fidelity, three-dimensional, cell-resolved simulations of the flow of deformable RBC suspension in two physiologically realistic microvasculatures obtained from published *in vivo* images [14,61]. We refer to them as vasculatures A and B, respectively (figure 1). They correspond to two different organs: network A represents an area of the mesenteric vasculature as given in [61], while network B represents the retinal vasculature as in [14]. Additional information about the structural parameters is provided in electronic supplementary material, figure S1. It should be noted that the distribution of vessel diameter over successive generations generally follows Horton's law [2] which provides some sort of universality of vascular structures at the smallest scales when the aggregate distribution is considered. However, when compared at the level of individual vessel/bifurcation, there are differences between the two networks as evident from electronic supplementary material, figure S1. Two vasculatures together are comprised more than 100 vessels, 42 bifurcations, and nearly 2000 RBCs flowing through them at any instant. Vessel diameters range from 5.5 to 24 μm . Simulation details are given in [32,42]. Briefly, fluid motion is governed by the unsteady Stokes equations and continuity equations that are solved using a projection-based method and finite-volume/spectral discretization. Flow in the networks is driven by specifying a flow rate boundary condition with a physiological flow rate of 1.2–1.3 nl s^{-1} specified in the main inlet. RBCs are injected at the inlet with an average haematocrit of 30%. Each RBC is modelled as a sack of fluid enclosed by a zero thickness hyperelastic membrane and having a biconcave discocyte resting shape with an end-to-end distance of 7.8 μm . The RBC membrane is assumed to possess a resistance against shearing, area dilation and bending. The viscosity of the haemoglobin and plasma is taken to be 0.006 and 0.0012 Pa-s, respectively. A two-way coupling between the fluid and cells is implemented via the continuous forcing immersed boundary method. Each RBC surface is discretized using 5120 triangular elements. As cells are deformed by the fluid, the elastic tension in the cell membrane develops which is obtained using a finite-element method, and then coupled back to the surrounding fluid. RBC–RBC aggregation is not considered. Immersed boundary method is also used to implement the boundary conditions on the vessel surfaces. About 160 million mesh points are used for each vasculature where three velocity components and pressure are obtained, and about 5 million mesh points are used where information about shape and membrane tension is stored for all cells. The physical flow time simulated is about 1.5 s. The velocity and pressure fields are stored every 5 ms, and the cell shapes are stored every 0.5 ms. The simulation results have been validated against *in vivo* data, and they accurately predict spatially and temporally varying blood flow and RBC distributions and other haemodynamic parameters in the vasculatures [42].

A visualization of RBC distribution at one time instant, and relevant time-series haemodynamic parameters, namely, the blood flow rate (Q) and RBC volume fraction or haematocrit (H), as predicted by the simulations are presented in figure 1. Here we will refer to these simulation data as 'direct simulation

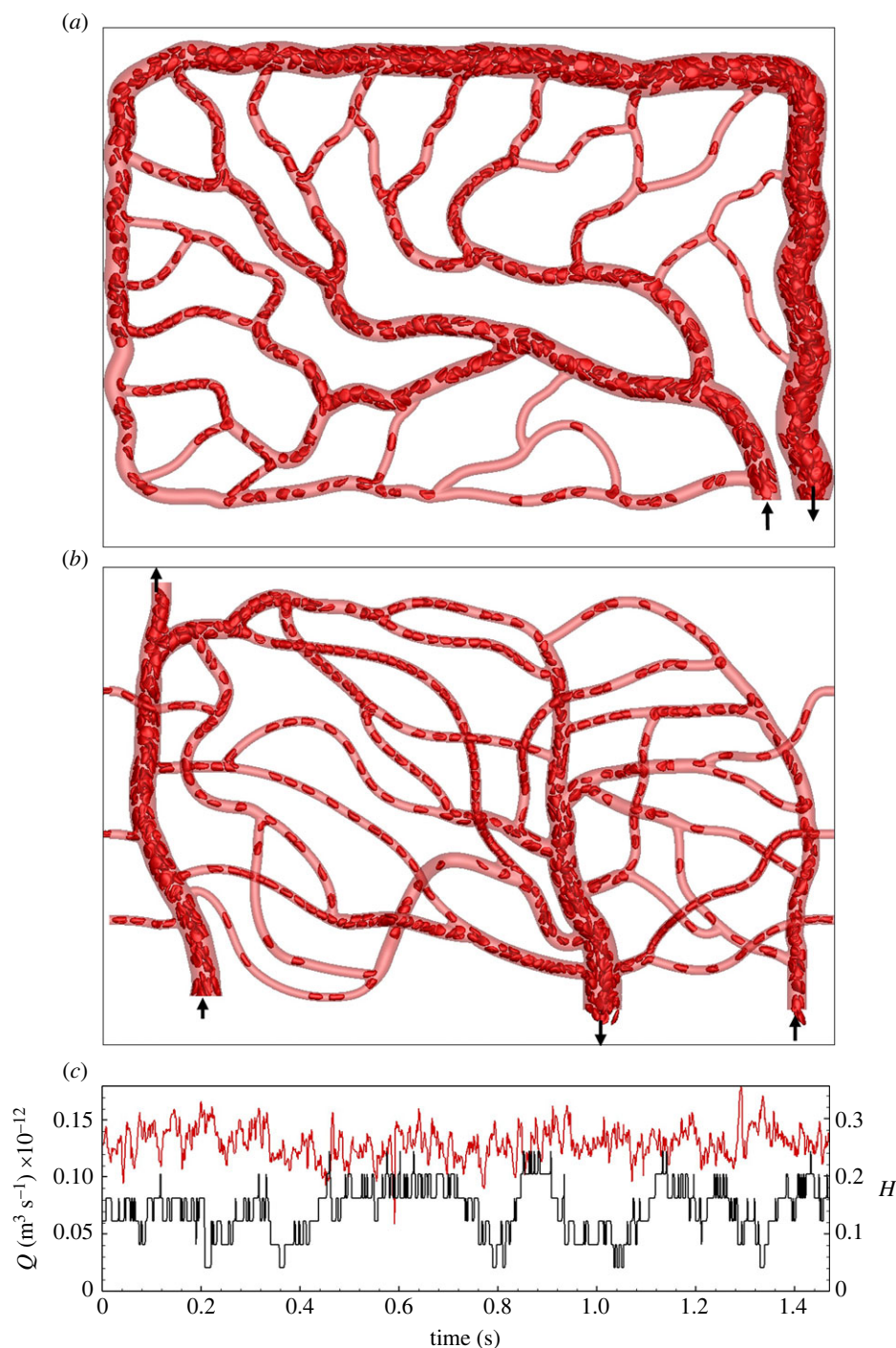


Figure 1. Data acquisition via *direct simulation results* (DSR). (a,b) High-fidelity cell-resolved simulations of RBC suspension flowing through vasculatures A (top) and B (middle) used to generate data for ML model training and validation. Visualization from one time instant is presented showing RBC distribution for each vasculature. Arrows indicate inlets/outlets. (c) DSR-predicted time-dependent flow rate ($\dot{Q}$; coloured red; left axis) and RBC volume fraction or haematocrit (H ; coloured black; right axis) in one vessel in vasculature B over more than one cardiac cycle is shown. Additional data are in the electronic supplementary material (#S6; figures S6–S8).

results' (DSR). The highly deformed shape of flowing RBCs, and a heterogeneous RBC distribution as typical of physiological scenarios, are predicted by the DSR as may be noted in figure 1a,b. Also, strong time-dependent variations in blood flow and haematocrit are observed that arise due to RBC–RBC and RBC–vessel interactions as shown figure 1c from our DSR.

2.2. ML models

Here we provide a brief overview of two specific ML algorithms used in this work. They fall under the broader category of supervised learning which is the model mostly used for haemodynamic problems. In this approach the ML model

builds a relationship $[Y_1, Y_2, \dots, Y_m] = F([X_1, X_2, \dots, X_n])$ where $[X_1, X_2, \dots, X_n]$ is the input data, $[Y_1, Y_2, \dots, Y_m]$ is the output and F is the mapping function. Depending on the type of output, there are two sub-categories within supervised learning: the classification model and the regression model. In the former, the input data are classified into different classes. The output is the probability of a given input belonging to one class. In the regression model the output is a real-valued continuous variable that is a measurable quantity. For our present work, the classification model is considered in identifying the path of an RBC as it flows through a vascular bifurcation. For all other predictions, namely, the time-averaged and time-dependent blood flow and haematocrit, the regression model is used. Below, we discuss two types of ML models that are used for this purpose.

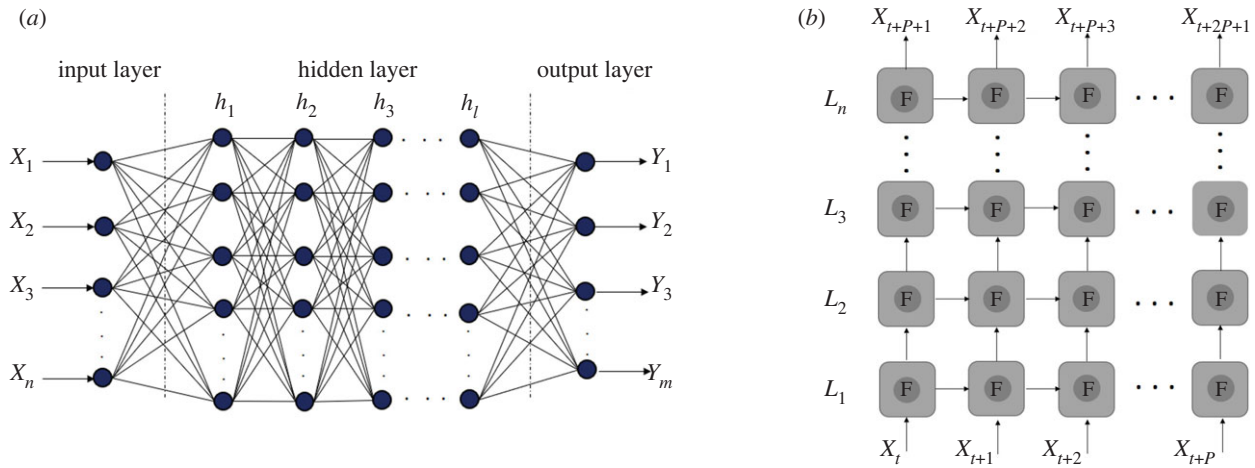


Figure 2. Schematic of (a) ANN and (b) multilayer long short-term memory (LSTM) models used for classification and time-average predictions, and for time-dependent predictions, respectively. For the LSTM model, input X_i represents a data in a time series between t and $t + P$, and output Y_i represents future prediction of X for time $t > t + P$.

2.2.1. Artificial neural network (ANN)

ANN is used in the current study for the classification problem and for the regression problem dealing with the prediction of time-averaged quantities. ANN is chosen because it is a robust method when a complex nonlinear relation exists between input and output, unlike linear ML models. Also, there is no limitation on the number, type (integer versus real) and range of values of inputs for ANN. Furthermore, ANN has the ability to handle the data in parallel which can be helpful to perform multiple tasks at one time. Figure 2a presents a general schematic of ANN. The inputs are passed through several computational units or nodes known as *neurons* that are arranged in *layers*. The first layer is the input layer, the last layer is the output layer and the intermediate layers are called *hidden layers*. The output of one layer serves as the input to the next. Each neuron in a hidden layer serves as one regression model, composed of its input data, *weights*, a *bias* and an output. The input to a neuron is calculated as the weighted sum of the values received from the neurons in the previous layer. Once the input is calculated, a *transfer* or *activation function* is used to obtain the output of that neuron, which is then passed to the next layer. The process is repeated until the final output of the ML model is obtained. Typical activation functions could be nonlinear, sigmoidal or hyperbolic-tangent, and their choice depends on the type of problem being addressed. For the regression problems, we use the *rectified linear activation function* (ReLU) which is $\mathcal{F}(\eta) = \max\{0, \eta\}$, where η is the input for a neuron. For the classification problem, we use ReLU for all hidden layers except the last one, whereas for the last hidden layer, we use the *softmax activation function* which maps the input data to an output value between 0 and 1 which is the probability of the data being in one of the multiple classes.

The goal of the ANN model is to obtain the weights and biases through the training process so that the difference between the output predicted by the ML model and DSR is minimized. The training process starts with the forward propagation using the first input data and random weights. Once an ML output is predicted, the *error* (or *loss function*) relative to the DSR is calculated. The weights and biases are then updated using, e.g. a gradient descent method, such that the loss function is minimized. Using the updated weights and biases, the next input is considered, and the process is repeated. Once the last input is used, an *epoch* is completed. At the end of an epoch, one achieves significantly improved values of the weights and biases. With these, another epoch begins, and the process is repeated for many epochs until the error is sufficiently reduced.

The accuracy of ANN models depends on a number of parameters called hyperparameters, such as, the number of hidden

layers, number of neurons in each layer, type of activation function and optimization, number of epochs, etc. Different hyperparameters for each specific problem considered here are mentioned in respective sections.

2.2.2. Long short-term memory

For the prediction of the time-dependent blood flow rate and haematocrit in each vessel, we use the long short-term memory (LSTM) model as shown in figure 2b, which is a commonly used type of recurrent neural network (RNN). While for some applications ANN may not be suitable to handle time-series data, RNNs are designed to deal with such data. RNN models use some prior inputs, e.g. the history in a time-dependent data, as well as the current input, to predict future data. One issue with RNNs is that if the time-series data have a dependency on an extended period of past data (vanishing-gradient issue), predictions may not be good. LSTM is designed to overcome this issue and can handle data dependency over a long time. Additionally, LSTM can take into account a nonlinear dependency between input and output. In the current study, we use a multilayer LSTM as shown in figure 2b for higher prediction accuracy.

Note that unlike ANN and LSTM, which are suitable to handle data as in the present study, convolutional neural network models (CNNs) are useful for processing images, and hence not used here.

3. Results

First, we consider the time-averaged but spatially varying results. This is followed by the prediction of the time-dependent (but spatially limited) quantities. Then, the prediction of simultaneous spatially and temporally varying quantities is given.

We use the DSR data from one microvasculature, say vasculature A, to train (build) the ML models, and another microvasculature (B, in this case) that is geometrically different for testing the model.

3.1. Time-average blood flow rate distribution

At the root of the problem lies building a relationship between the flow rate in a mother vessel and that in a daughter vessel of any bifurcation in a vasculature, given certain geometric information as the input to the ML model as schematically presented in figure 3a,b. For a vasculature consisting of N_B

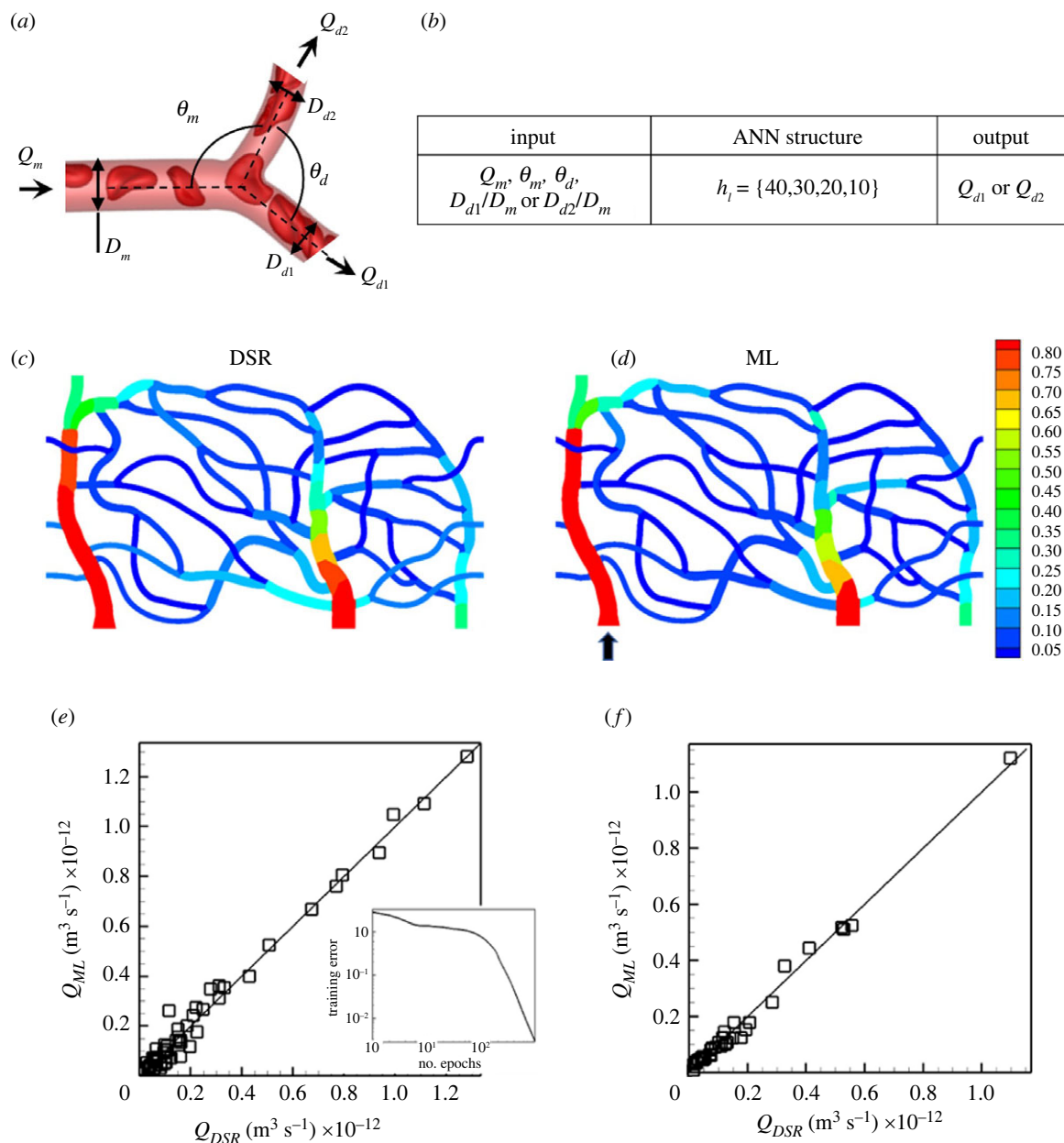


Figure 3. Prediction of time-averaged blood flow rate in a vasculature. (a) Model schematic showing a vascular bifurcation with input and output features. Q_m , Q_{d1} , Q_{d2} are flow rates in the mother and two daughter vessels, respectively, D_m , D_{d1} , D_{d2} are their diameters, and θ_m and θ_d are angles between the vessels as marked. (b) Table showing the inputs, output, and ANN structure. (c,d) Maps of time-averaged flow rates in vasculature B obtained from DSR and predicted by the ML model that is trained using the data of vasculature A. Blood flow rate (in $\text{m}^3 \text{s}^{-1} \times 10^{-12}$) in the entire vasculature is predicted for a specified flow at the main inlet as marked by an arrow. Contours are from 0.05 to 0.8 with an increment of 0.05. (e,f) Quantitative comparison of ML prediction (Q_{ML}) against DSR (Q_{DSR}). Each data point represents the flow rate in an individual vessel of a vasculature. For (e), vasculature A is used for training and B is used for prediction. For (f), vasculatures are swapped for training and prediction. Inset in (e) shows how training error decreases with an increasing number of epochs.

number of bifurcations, the input consists of $2N_B$ vectors, each of which contains four input features, namely, the flow rate of the mother vessel, the ratio of either one of the chosen daughter vessel diameters to the mother vessel diameter, the angle between two daughter branches, and the angle between one daughter vessel and the mother vessel. The output is $2N_B$ vectors representing the flow rates in $2N_B$ daughter vessels. Note that a single ML model is built that takes into consideration all bifurcations and vessels in the vasculature.

Note that the ML models for flow rate do not consider vessel haematocrit. The presence of the cells affects the flow rates but that is already considered by the DSR data. Also, the training and testing networks need not have same number of bifurcations or vessels.

We use an ANN with four hidden layers having 40, 30, 20 and 10 neurons, consecutively. This architecture is obtained via numerical experiments and found to provide satisfactory results (see electronic supplementary material, figure S2). The ML model is trained using the DSR time-averaged flow rates for either of the vasculatures A and B. Convergence of the model is determined by mean squared error between the ML-predicted output and the DSR for the training vasculature. The inset in figure 3e shows that the error decreases with a sufficient number of epochs, yielding an aggregate training accuracy above 98%.

Once the ML model is obtained for, say, the vasculature A, we use this model to predict the blood flow in vasculature B. For the prediction, we start with the flow in the main inlet

to the vasculature B as known, and sequentially proceed to predict the flow in subsequent branches.

Figure 3c shows a map of the time averaged blood flow as obtained by DSR in vasculature B, and figure 3d shows the same quantity as predicted by the ML model based on the training using vasculature A. The agreement between the two cases is quite good.

Figure 3e and f present the ML-predicted flow rates (Q_{ML}) compared against the DSR data (Q_{DSR}). In figure 3e, vasculature A is used for training and B for prediction, and vice versa in figure 3f. Agreement between the ML prediction and DSR is reasonable, with 82% aggregate accuracy. As in any ML problem, the accuracy of the current prediction can be further improved by using a larger input dataset; our current model has $2N_B = 42$ datasets which we deem to be modest. Consideration of larger datasets will also reduce the extreme difference between geometric features of the training and testing vasculatures.

3.2. Prediction of red blood cell partitioning and time-average haematocrit distribution

Next, we predict time-averaged RBC distribution in a vasculature. For this, we first need to predict RBC partitioning at each bifurcation of the vasculature of interest. When RBCs flow through a bifurcation, some of them enter one daughter vessel while others enter the other vessel. Here we first build an ML model that would predict whether an individual RBC at any bifurcation would enter one daughter vessel or the other.

This ML model is formulated as a classification problem, and we use the ANN model with six hidden layers. The model is described in figure 4a,b. An ML model is built for each individual bifurcation in the training vasculature, thereby yielding N_B number of models for N_B bifurcations. The output of each model is a vector of length two which represents the probabilities of an RBC to enter either of the daughter vessels in the bifurcation being modelled. Whether the RBC enters one or the other daughter vessel depends on the geometry of the bifurcation, the flow rate, and the radial location of the RBC in the mother vessel. Accordingly, the input vector contains the vessel diameter ratios (D_{d1}/D_m , D_{d2}/D_m), flow rate ratios (Q_{d1}/Q_m , Q_{d2}/Q_m), the radial distance of the RBC from the centre of the mother vessel (r), and whether the cell is located on the side of the mother vessel that is closer to one daughter vessel or the other which is represented as a vector using binary numbers as $\{1, 0\}$, or $\{0, 1\}$. It may be noted that the use of two vectors of length two seems redundant since a cell can only enter daughter vessel. However, the classification model used here is a generic one that can address problems involving multiple classes. In the present context, this situation may arise when more than two daughter vessels emanate from one junction as sometimes observed in angiogenic vessels. One vasculature is chosen as the training vasculature, and all N_B models are trained using the DSR of that vasculature. For each individual bifurcation, the DSR provides the number of cells N_m in the mother vessels and the radial location of each cell $r_{i=1,N_m}$. The model then predicts how many of them enter one daughter vessel. Convergence of the training process is based on the cross-entropy loss defined as $-(1/n_c) \sum_{n_c} \log(P_i)$ where $n_c = 2$, for two daughter vessels, and P_i is probability of an RBC to enter a daughter vessel. Note that for DSR, $P_i = 1$. The cross-entropy loss is

presented in the inset of figure 4c for a randomly selected bifurcation and shows convergence with an increasing number of epochs.

Several articles have shown that events at upstream bifurcations affect partitioning [30,32,33]. However, this is implicit in the DSR data which are used to build the ML models, and hence one may claim that to a certain degree the ML models already include this effect. One can also explicitly include these in the ML models, but that will increase model complexity.

Before we use the model built using one vasculature to predict RBC partitioning for bifurcations in the other vasculature, we first *evaluate* its capability in the same training vasculature. This *evaluation* phase is different from the training phase as follows. Unlike in the training phase where each individual bifurcation is considered in isolation using the exact data of the DSR (N_m and $r_{i=1,N_m}$), now in the evaluation stage we *randomly* inject RBCs at the main inlet of the *training* vasculature and progressively predict the number of cells in each vessel as they flow through the spatial hierarchy of large arteries to the smallest capillaries. Thus, the number of RBCs upstream of each bifurcation and their radial locations differ from the DSR. Only N_m specified at the vasculature main inlet is matched to the DSR.

Figure 4c presents ML-predicted number of cells (N_{ML}) as obtained in the evaluation stage against the DSR data (N_{DSR}). The bottom inset in the figure shows the accuracy of the results for each vessel defined as N_{ML}/N_{DSR} or N_{DSR}/N_{ML} . The accuracy of the models is 72–97%, with the lower accuracy occurring in smaller capillary vessels and higher accuracy in larger vessels. The accuracy generally shows a decreasing trend with decreasing vessel diameter as shown in figure 4c (bottom inset) and in the electronic supplementary material (#S4, figure S4). This is because the smaller bifurcations feeding these vessels receive a smaller number of cells; as a result, the amount of training data is less. Secondly, RBCs exhibit extreme deformation and complex dynamics at such bifurcations. Such deformation and interactions become more complex at some bifurcations making them prone to larger errors. More complex ML models are needed to include such complexity.

We then use the models built from one vasculature to predict the RBC partitioning in the *other* vasculature; we term this the *prediction phase*. RBCs are randomly injected at the inlet of the *other* vasculature, and we progressively predict the number of cells in each vessel as they flow through the hierarchy of large arteries to the smallest capillaries. Since the training and prediction vasculatures are geometrically different, for each bifurcation a specific ML model is selected from the entire set of training models by matching the flow rate in the mother vessel and the geometry of the bifurcation. See electronic supplementary material (#S3) for further details. Figure 4d compares the ML prediction and the DSR for the number of cells in each vessel. As before, higher accuracy is obtained in larger vessels, and the mean accuracy is 79%. This is less than the evaluation accuracy because of the differences in geometry and flow rates of the two vasculatures, and a moderate number of training bifurcations. Figure 4e,f compares the maps of cell distribution in vasculature B as obtained from the DSR (figure 4e) and from the ML prediction (figure 4f) based on the training using vasculature A, which shows an overall good agreement between them.

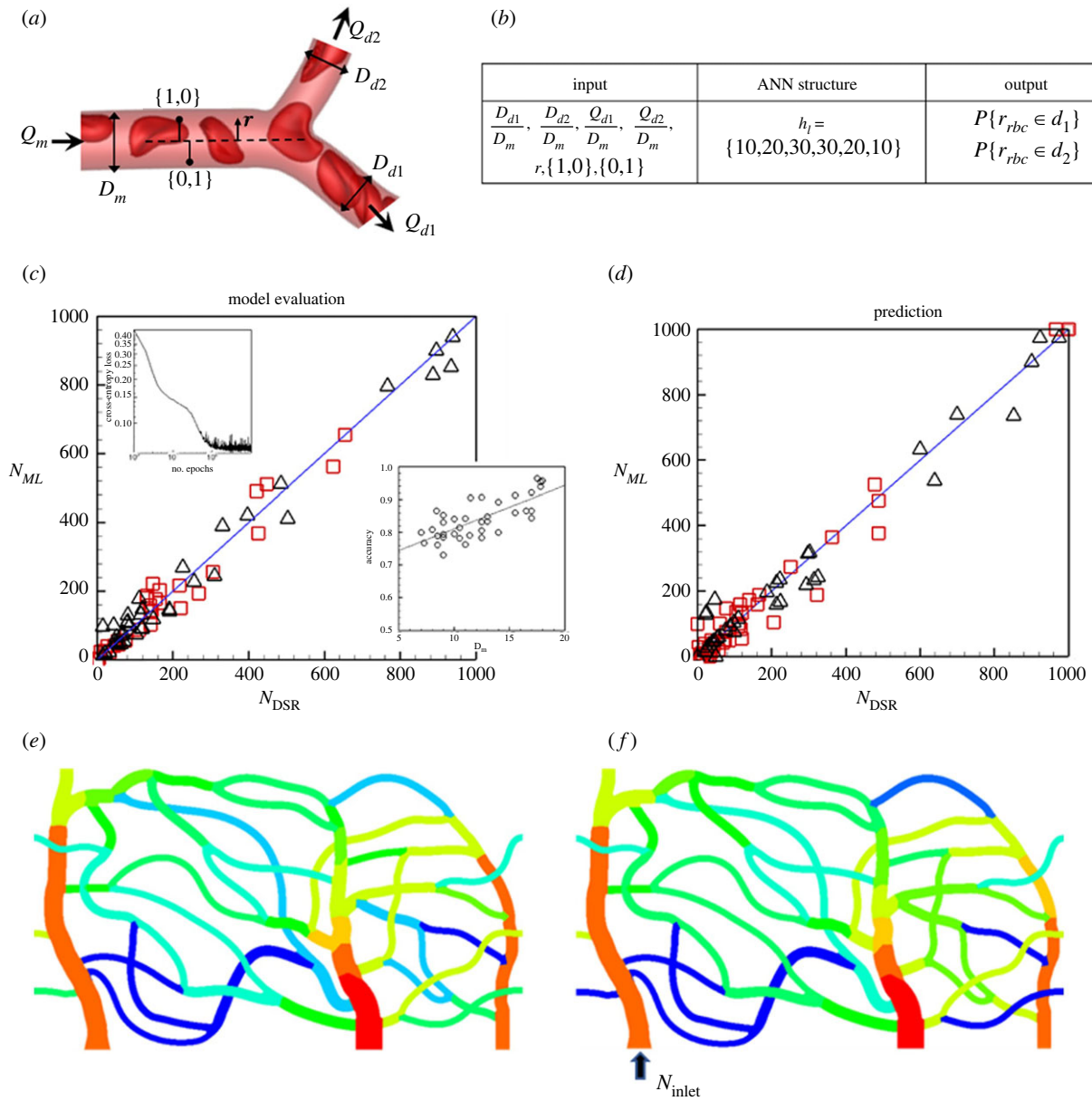


Figure 4. Prediction of RBC partitioning at bifurcations and cell distribution in the vasculature. (a,b) Model schematic and parameters (see text for definitions). (c) Results of the *evaluation* phase. The number of cells as predicted by the ML model (N_{ML}) in the evaluation phase is compared against the DSR. Each data point is the result for each vessel in the vasculature in consideration. Black symbols are for vasculature A, and red for vasculature B. The top inset shows convergence in terms cross-entropy error; the bottom inset shows the evaluation error (see text for definition) as a function of vessel diameter. (d) Results for the *prediction* phase. In this, the ML models based on vasculature A are used to predict vasculature B (black symbols), and vice versa (red symbols). (e,f) Maps of cell distribution in vasculature B as obtained from DSR (e), and from ML prediction (f) using the models built from vasculature A. Number of cells at the main inlets is specified and the ML models progressively predict cell distribution as it develops through the vessel hierarchy. An exponential scale is used for the contours ranging from 10 (blue) to 1000 (red).

Having predicted the RBC partitioning in the entire vasculature, we can now predict the time-averaged haematocrit distribution. For this, the haematocrit at the main inlet to the vasculature is first specified, and the calculation proceeds spatially through the hierarchy of bifurcations using the relation

$$H_{d1} = H_m \left(\frac{Q_m}{Q_{d1}} \right) \left(\frac{N_{d1}}{N_m} \right),$$

where H_m and N_m are haematocrit and number of cells in the mother vessel of a bifurcation, and H_{d1} and N_{d1} are those quantities for one of the daughter vessels. Figure 5a and b show quantitative comparison of the ML-predicted haematocrit versus the DSR in each vessel of both

vasculatures during the evaluation stage and prediction, respectively. Mean accuracy is approximately 85% and 82%, respectively. Figure 5c,d compares the maps of haematocrit distribution in vasculature B as obtained from the DSR (a), and predicted by the ML model (b) built using vasculature A. Note that the contours in figure 5 are drawn with an increment of 0.02, which makes the two plots look different. We used such small increment to highlight the difference between the DSR and ML prediction. Also note that ML is not expected to give back the exact result at every bifurcation. It provides an approximate prediction which always has some difference against the actual result. While judging the accuracy of the method, one should consider the aggregate accuracy over the entire network rather than individual vessel/bifurcations.

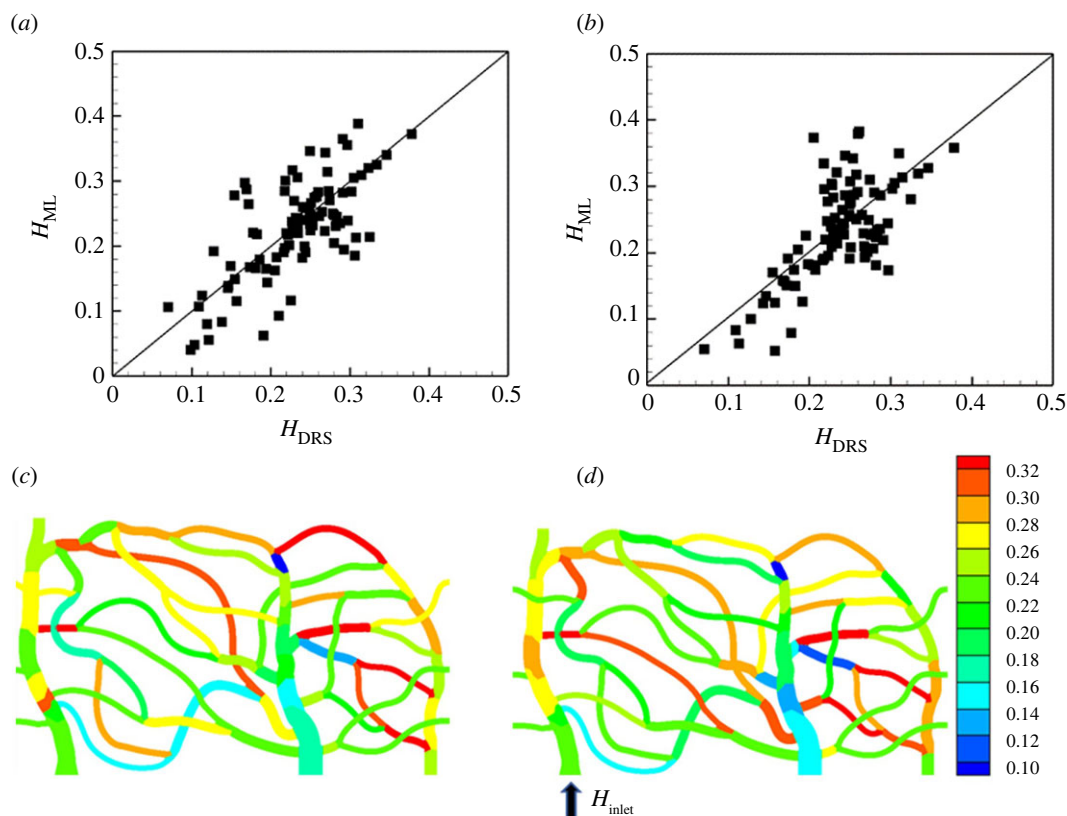


Figure 5. Haematocrit prediction. Comparison of the ML model and DSR results for (a) the evaluation phase, and (b) the prediction phase. Each data represents haematocrit in each vessel of the vasculatures. (c,d) Comparison of haematocrit maps in vasculature B as obtained from the DSR (c) and predicted by the ML models built from vasculature A. The haematocrit at the main inlet (black arrow) is specified, and the models progressively predicts through the hierarchy of vessels.

3.3. Prediction of time-dependent haemodynamics

In general, haemodynamic quantities in a microvessel are unsteady in nature, and their fluctuation frequency spans over a wide range. Under physiological conditions, low-frequency fluctuations may arise from factors such as pulsation of the heart, vessel contraction, and capillary blockage by leucocytes. High-frequency oscillations occur due to RBC–RBC and RBC–vessel interactions, and RBC jamming at bifurcations. At times, these latter factors can also cause low-frequency oscillations. In our DSR, we do not model flow pulsation, vessel contraction or leucocyte movement. Thus, haemodynamic quantities in our DSR fluctuates due to RBC effects. The objective of this section is to describe the framework of an ML model that can predict such time-dependent variations. First, we consider each vessel *separately* without considering its connections with surrounding vessels or junctions in the vascular hierarchy, so that the problem is purely a *time-dependent problem*. The full problem of space and time-dependency is considered in §3.4.

Specifically, the time-dependent problem concerned here is about forecasting the flow rate Q and haematocrit H in a vessel for time $t > T$, given that the DSR data are available for $t \leq T$.

For this, we employ a four-layer stacked LSTM. To train the ML model, first we need to construct input ($X^{(DSR)}$) and output ($Y^{(DSR)}$) vectors from the DSR data. The input vector is the time dependent DSR data over a window of time t and $t + w$, and the output vector is the data over a window $t + w$ and $t + 2w$, with $t < T$ being a sliding time (figure 6a for a representative case). Note that the DSR data are discrete data at every 0.5 ms. For an input $X^{(DSR)}$, the LSTM model gives an output $Y^{(ML)}$ at every epoch. The convergence of

the model is achieved when the mean-squared error between $Y^{(DSR)}$ and $Y^{(ML)}$ decreases sufficiently over several epochs.

Our numerical experiments show that the accuracy of training depends on the vector length which in this case is the time window size w . A large window cannot capture the high-frequency oscillations, while a small w will produce significant noise. To determine the optimal w , we define an objective function using three moments of the time-series data as

$$f = a |\sigma_{ML} - \sigma_{DSR}| + b |S_{ML} - S_{DSR}| + c |K_{ML} - K_{DSR}|,$$

where σ , S and K are standard deviation, skewness, and kurtosis, respectively, and the coefficients $a \approx \mathcal{O}(10^2)$, $b \approx c \approx \mathcal{O}(1)$ are determined by minimizing f . Multiple runs are made with different window sizes, and for each of them convergence is achieved. The optimal window is then chosen from them and found to be 0.175–0.225 s for most vessels. See electronic supplementary material, figure S5 for further information.

Once an ML model is trained for a selected vessel, it is used to forecast the haemodynamic quantities in future time $t > T$. The same optimal window is used as obtained from the training process, and the prediction progresses in time. At any step, the input is the data between t_m and $t_m + w$, and the output is for $t_m + w$ and $t_m + 2w$, where $t_m \geq T - w$ (figure 6a). Thus, the output of one step is the input to the next. Figure 6a,b compares the DSR data for $t < 1.5$ s and ML prediction for $1.5 < t < 5$ s for one selected vessel and shows that our model can capture the time variations of Q and H quite well.

In figure 6c,d we compare the mean, σ , S and K , as obtained from the DSR and predicted by our ML model for

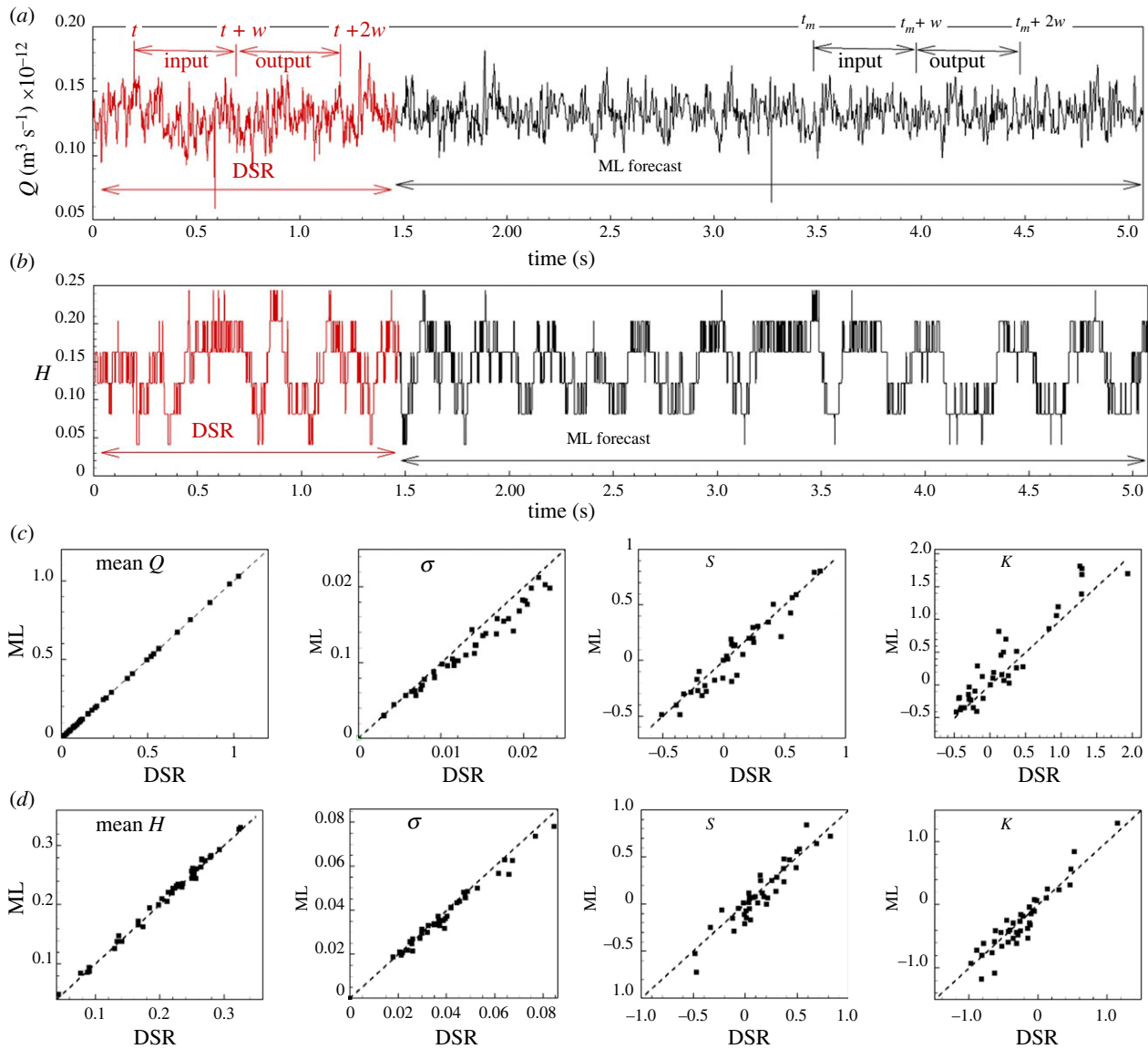


Figure 6. ML prediction for time-dependent haemodynamics in individual vessels without considering their hierarchical location in the vasculature. Predictions of blood flow rate (Q , a) and haematocrit (H , b) are shown for one representative vessel. Time-series DSR data which is available for $t \leq 1.5$ s is shown in red, and the ML prediction which is for $t \geq 1.5$ s is shown in black. (c, d) Accuracy of the time-series prediction is presented by comparing the mean, standard deviation (σ), skewness (S), and kurtosis (K) as obtained from the DSR and ML forecast. Data for additional vessels are given in the electronic supplementary material (#S8, figures F9 and S10).

all vessels in the two vasculatures. It shows that the ML predictions agree well with the DSR.

3.4. Prediction of simultaneous spatial and temporal variations

Next, we predict the evolution of haemodynamic quantities simultaneously in time and space in a vasculature. As blood and RBCs flowing through a vessel encounter a bifurcation, the way they partition into the daughter vessels determines their spatial evolution in the microvasculature. The time-dependent variations of such quantities in the daughter vessels, however, differ from those in the mother vessel based on the location of the bifurcation in the vasculature, and due to RBC–RBC and RBC–vessel interactions. We first build ML models which predict the time-dependent flow rates and haematocrits in the daughter vessels of a single *isolated* bifurcation by knowing such quantities in the mother vessel as the input.

The ML model for each bifurcation is based on the four-layer stacked LSTM and consists of a time-dependent quantity of the mother vessel obtained from the DSR as the input, and two time series data in the daughter vessels as outputs. The training is performed using the optimum time window as obtained in §3.3. Once we train the models, we test their predictability as follows. For any bifurcation, we supply the DSR + *forecasted* time series for the mother vessel as obtained from §3.3. The model then forecasts the corresponding time series in the two daughter vessels of that bifurcation. Results from one such prediction are given in figure 7 and compared against the DSR. Accuracy is evaluated in terms of the mean and three moments (σ , S , K). The DSR data are for $t < 1.5$ s, and the ML predictions in the daughter vessels are for $0 < t \leq 5$ s. The DSR data almost overlap with the ML prediction as seen in the figure.

We build such ML models for space time predictions for all bifurcations (in isolation) in vasculature A, and then use them as a model bank to predict simultaneous space time evolution in connected bifurcations and vessels in vasculature B. Selection of a particular ML model out of the model bank

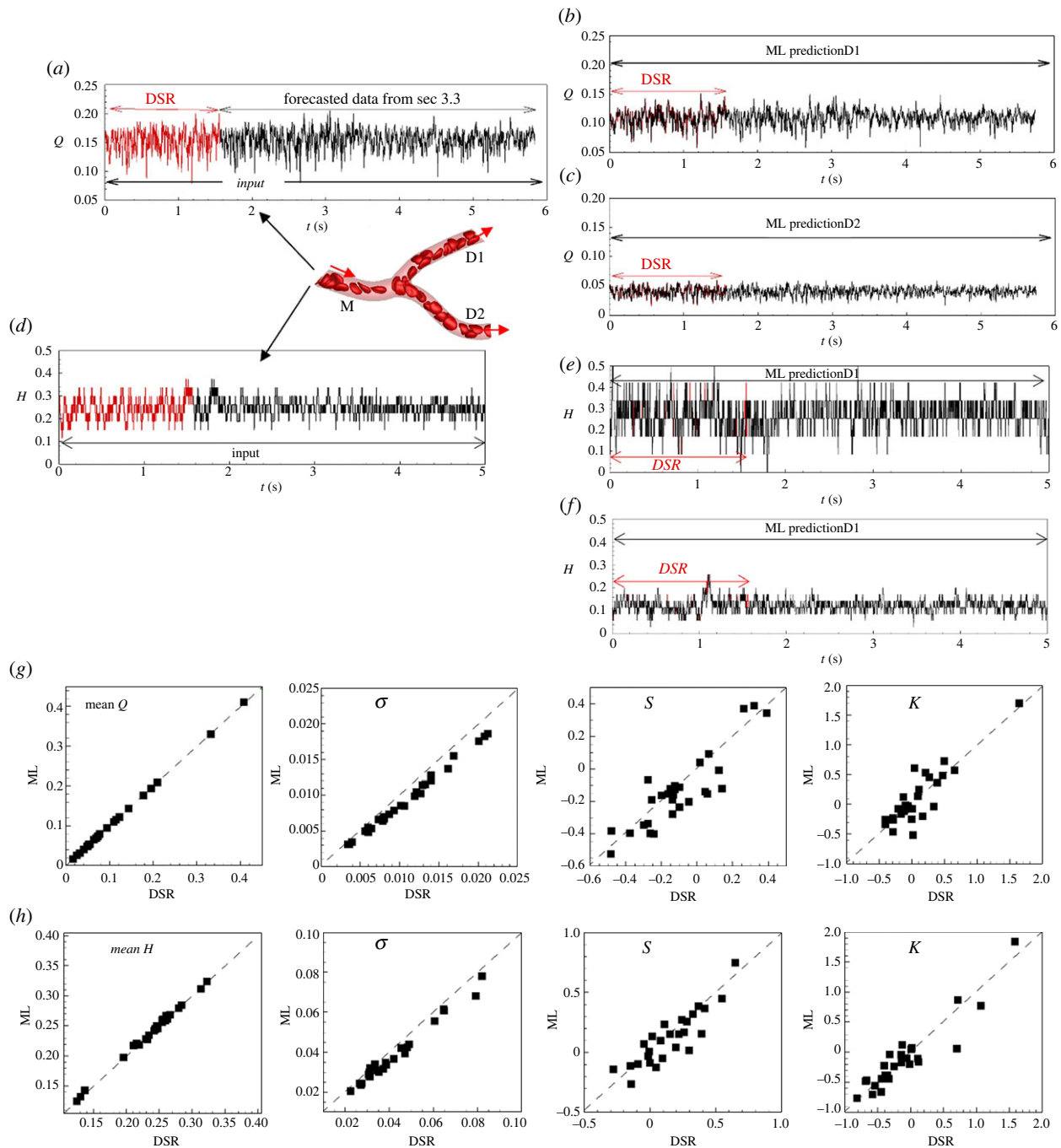


Figure 7. ML prediction of time- and space-dependent quantities in daughter vessels with given data in the mother vessel of a bifurcation. (a,b,c) shows blood flow rates in the mother vessel (M; input), and predicted flow rates in the daughter vessels (D1, D2). ML predictions (coloured black) are compared against DSR data of the daughter vessels (coloured red). Note that the DSR data are for $t < 1.5$ s, and the ML predictions in the daughter vessels are for $0 < t \leq 5$ s. The DSR data almost overlap with the ML prediction. (d,e,f) Haematocrit prediction plotted in the same manner. (g,h) Accuracy of time-series predictions is shown by comparing the mean, standard deviation (σ), skewness (S), and kurtosis (K) as obtained from the DSR and ML forecast.

that is most suitable for a specific bifurcation in vasculature B is done by matching, as close as possible, the three time-averaged quantities, namely, the blood flow rate and haematocrit in the mother vessel, and the ratio of the flow rates in the mother to a daughter vessel. Figure 8 shows a representative result where the time-dependent flow rate and haematocrit in one vessel are used as inputs, and those in a vessel that is located two bifurcations downstream are predicted. The inputs are constructed using the DSR data for $t < 1.5$ s plus the time-dependent forecast for $1.5 < t \leq 5$ s as obtained from §3.3. The ML predictions are for $0 < t \leq 5$ s. The ML predictions are compared against the DSR data available for $t < 1.5$ s and seen to agree well.

4. Discussions and conclusion

We investigated the applicability of the ML techniques for the prediction of time-averaged and time-dependent blood flow rate and RBC distributions in physiologically realistic capillary vessel networks. To train and test the models, we acquire data from high-fidelity simulations of the flow of deformable RBC suspensions in the networks. We use two networks that are geometrically quite different, with one used for the model building and the other for prediction. For the prediction of time-averaged blood flow rate, a regression-type ANN model is used, while for the prediction of time-average RBC distribution, a classification-type ANN

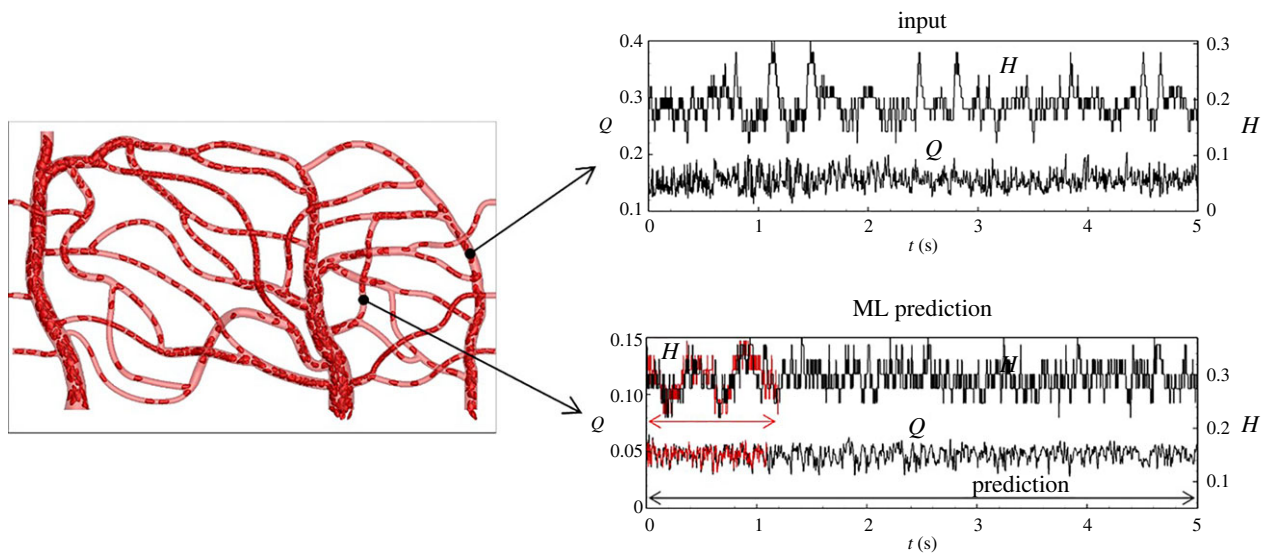


Figure 8. ML prediction of simultaneous spatial and temporal evolution of blood flow rate and haematocrit in a vessel that is located two bifurcations downstream from the input vessel. Vasculature B is predicted using ML models from vasculature A.

model is used. With the flow rate and haematocrit specified at an inlet of vasculature, the models predict time-averaged flow rate and RBC distributions in the entire network. For the prediction of time-dependent quantities, we use LSTM models. With these, we first predict the time-dependent flow rate and haematocrit in each vessel in isolation over a long time, as well as such quantities in isolated vascular bifurcations. These models constitute a model bank which is then used to predict simultaneous spatially and temporally evolving quantities through the vessel hierarchy in the networks. We also find that an optimal vector length (or time window) is needed to predict fluctuations in the time-series data.

While the accuracy of the training is above 90%, that of the prediction is around 80%. As in any ML application, accuracy improves as more data are used in training. Each vasculature considered here has about 21 bifurcations which are deemed to be modest. Furthermore, two vasculatures differ from each other geometrically as well as in terms of blood flow and RBC distributions. The accuracy of the prediction depends on how closely the parameters of the target vasculature, which include the geometric features of the bifurcations, the flow rate, haematocrit and flow rate ratios, match with those of the training vasculature. Availability of additional training data spanning over a larger parameter space will certainly improve the prediction accuracy. One can also perform simulations using synthetic isolated bifurcations to generate data over a larger parameter space. The simulation tool to generate DSR data runs on high-performance platforms. Other simulation tools, e.g. the HEMOCCELL system, <https://www.hemocell.eu>, also exist that can run on high-performance computers and such tools can be used to generate much more DSR data over longer times and could enable us to further improve the ML models.

Additionally, we find that both training and prediction accuracies are higher for vascular junctions with larger diameters, but they decrease as capillaries with small diameter are approached. This is because cell–cell and cell–vessel interactions become the dominant mechanism of cell partitioning at such small bifurcations. This is also the reason why lower accuracy occurs in predicting RBC distribution

than flow distribution. More complex ML models will be needed to account for such interactions to improve the accuracy. However, these smaller capillaries are often the terminal vessels, and therefore the reduced accuracy in such vessels is not expected to significantly alter the RBC distribution in the entire vasculature. Indeed, the comparison of flow rate and haematocrit maps in the entire vasculature obtained from the DSR and by the ML models shows a good agreement between the two at the network scale. Furthermore, since these errors primarily arise due to the limited amount of training data and lack of more detailed models for complex RBC–RBC and RBC–vessel interactions at individual bifurcations, it cannot be said that they continually *accumulate* over successive bifurcations, and we have observed cancellation of errors. Our accuracy analysis supports this.

Despite the limited amount of training data and the absence of explicit incorporation of each RBC's dynamics, the results from the current ML models are promising. To our knowledge, this is the first study to use ML models for physiologically realistic microvascular networks. This technique can be applied to new applications in the future. For example, the models may be extended to consider a much larger and organ-scale microvasculature over a very long time without the need for an expensive high-fidelity computation. Experimental data can also be used instead of computational data. Furthermore, the ML models can be used for the prediction of haemodynamics in blood cell disorders, such as sickle cell anaemia, malaria, and diabetes mellitus which are characterized by reduced RBC deformability, provided that the training data are available. Additionally, the model can also be extended to predict altered haemodynamics during vascular adaptation. In this study, our intention was to see if ML can be used to replace expensive simulations. Our intention was not to use ML to learn new physical laws. Indeed, ML can be (and has been recently) used to discern new physical laws (e.g. [59]). It can also be used to do so for RBC flow dynamics in capillary networks which is something that we intend to address in future. Our focus here is only on predicting flow rate and RBC distribution. For any other parameters, such as wall shear

stress, one can similarly use DSR data and develop ML models specific to such haemodynamic parameters. If this approach is taken, then one can expect an accuracy in the same range given the similar amount of training data used here. Alternatively, from our predicted flow rate, one can get wall shear stress by making the assumption that the velocity profile across the vessels is given by the Poiseuille's law (and, hence, blood behaves as a Newtonian fluid), $\tau = -4\mu Q / \pi R^3$. Of course, the assumption is not valid for blood flow in capillary vessels and will introduce additional errors. Furthermore, the ML models presented here are not designed to predict velocity and haematocrit profiles across vessels. For that, different ML models need to be built as that would be a problem of higher degree. Moreover, in realistic and more complex vascular topologies, not all values can be measured, or information can be missing for some vessels. This issue is reflected in the current approach wherein the information is 'missing' for network B except one or two

vessels. Robustness of the approach is evident by the number of vessels, bifurcations predicted and the accuracy.

Data accessibility. ML codes and data are provided in https://github.com/SamEbiRutgers/ML_Microvasculature.git. The paper is based on data published in [32]. This is open-access and hence publicly available.

Electronic supplementary material is available online at [62].

Authors' contributions. S.E.: data curation, formal analysis, investigation, methodology, writing—original draft, writing—review and editing; P.B.: conceptualization, funding acquisition, supervision, writing—original draft, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interest declaration. We declare we have no competing interests.

Funding. This work was supported by a grant from the National Institute of Health (R01EY033003).

Acknowledgement. Computational resources at Pittsburgh Supercomputing Center, Texas Advanced Computing Center and Rutgers University are acknowledged.

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